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(1) Research Program/Project/Study Title: Development of Shrimp (<i>Metapenaeus ensis</i>) Ice Cream	
Proponent Name:	Stephanie S. Pimentel, Ph.D.
Academic Rank:	
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Telephone/Cell Phone/Fax Number:	
E-mail Address:	

Program or Project has component (if no proceed to 2)? ☐ Yes ☐ No

Component 1 Title:	
Component Leader/s:	
Academic Rank:	

(2) Category (please check) ☐ Social Science/Education /Humanities & other related fields ☐ Natural/Biological Sciences ▪ Food Research ☐ Developmental/ Computer/Engineering/ Architecture ☐ Extension		(3) Research Center [Check the concerned center] ☐ SSITRDC Social Science and Industrial Technology Research Development Center ☐ LRDC Livestock Research Development Center ▪ FFRDC Fisheries and Food Research Development Center ☐ CSRDC Crop Science Research Development Center
(4) Site of implementation:		
Municipality/City:		Roxas City
Province:		Capiz
Region:		VI
(5) Classification Research: Development: ▪ Basic ☐ Pilot Testing ☐ Applied ☐ Tech. Promotion/Commercialization	(6) Mode of Implementation ☐ Single Authorship/Proponent ▪ Collaborative Research (within campus) ☐ Collaborative Research (intercampus) ☐ Collaborative Research (external agency)	

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(7) Sector/Commodity Shrimp Ice Cream	(8) Discipline/Field of Specialization Food and Technology
(9) Rationale Ice cream is the most known dairy dessert, it is considered a nutritive food because it is mainly made of milk and fruits, representing a source of proteins, some vitamins, minerals, and some phytochemicals (<i>Balthazar, C. F. et al.</i>) Ice cream has elements that help strengthen your immune system. Milk contains vitamins and nutrients including vitamin A, vitamin D, selenium, and zinc which are all essential for a healthy immune system. Ice cream can also make us happier and can satisfy pregnancy cravings, it is manufactured from milk, which is packed with iodine, a vitamin that is crucial for nursing mothers and pregnant women. Additionally, iodine is crucial for the development of the fetal brain. <i>Metapenaeus ensis</i> sometimes called the greasyback shrimp or sand shrimp is a species of prawn. <i>Metapenaeus ensis</i> is a common local shrimp of commercial importance especially in Panay Island. The nutritional profile of Shrimp is excellent. Shrimp is a good source of protein and is also high in a variety of vitamins and minerals. Since prawns include omega-3 fatty acids and the antioxidant astaxanthin, eating prawns may also be good for your heart and brain. Ice cream with <i>greasy back shrimp</i> is an innovative and unique product. The sweet flavor of the ice cream and the salty flavor of the shrimp blended together on the tongue flawlessly. It could potentially be possible to improve the ice cream's nutritional and functional value by using these flavors.  <i>Metapenaeus ensis</i> (De Haan, 1844) (9-A) Project Description (<i>Discuss the project strategy and plan of action of the project. It should also describe the technical procedures and other work needed to accomplish the project objectives. The relationship between the project inputs and activities should be explained clearly in this section. Projects similar/related to the proposed project should be mentioned.</i>) The research will provide the composition and evaluation of a Shrimp Ice Cream composed of the following ingredients and processes; Shrimp is the main ingredient which is locally available and rich in nutritional factors. Other ingredients such as coconut milk, sugar, arrowroot, fresh milk, all-purpose cream, salt will be used. The acceptable of Shrimp Ice Cream will be based on sensory qualities such as taste, texture, aroma, color, and general acceptability. The best formulation will be used and undergo for consumer acceptability. Completely Randomized Design will be used with a single factorial design with four (4) treatments and three (3) replications. Analysis of Variance will be the statistical tool to use. This will be conducted at the FFRDC laboratory for 5 months including the Preparation of materials, product formulation and evaluation, statistical analysis, consumer acceptability, and write-ups.	
(10) Objectives/Statement of the Problem This study aims to, 1. Determine the level of acceptability of Shrimp Ice Cream; 2. Determine the sensory properties of the Shrimp Ice Cream in terms of taste, texture, aroma, color, and general acceptability;	

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3. Determine the microbial properties of Shrimp Ice Cream; and
4. Determine the consumers' acceptability of Shrimp Ice Cream.

(11) Scientific Basis/Theoretical Framework (Include concepts, theories and variables that indicate relevant relationships to be tested.)

INPUT	PROCESS	OUTPUT
Shrimp (<i>Metapenaeus ensis</i>) and other ingredients.	- Production and formulation of Shrimp Ice Cream - Consumer Testing and sensory evaluation	- Optimized formulation of Shrimp Ice cream - microbial properties product

(12) Review of Literature/Prior Art Search (Related researchers/activities that have been conducted. State-of-the-art of current technology/ information from which the project proposal will take off).

No other dish compares to ice cream for its ability to satisfy our appetites for sweetness. A less popular and unusual ice cream flavor is seafood, which has a distinctive flavor. It might contain seasonings, cream, and ingredients like lobster, crab, or even prawns. A few daring chefs and ice cream producers have experimented with savory seafood-infused ice creams for a distinctive gastronomic experience, despite the fact that it is not readily accessible.

The main ingredients of ice cream are milk, cream, sweeteners, natural flavorings, and other optional ingredients such as eggs, nuts, fruits, chocolate, and candy. There are various types of ice creams available in the market including a wide range of flavors, colors, textures, and ingredients. Both the flavorings and the sweeteners are imparting sweet sensation to the ice cream. (Perera,2021)

CN104286359A Production method of seafood ice cream

Abstract; The invention provides a production method for seafood ice cream. The production method comprises the following steps: cooking 20-25 parts of milk in a pot; putting the cooked milk into an ice cream mold; adding 10-15 parts of seafood stock and 6-9 parts of table salt; putting the mold into a refrigerator and freezing for 60-70 minutes; and then taking out to obtain the seafood ice cream. The production method of seafood ice cream has the advantage that the prepared ice cream has a unique taste and has the effects of relieving summer heat and cooling.

CN103859130A Shrimp-protein-containing ice cream and preparation method thereof

The invention provides shrimp-protein-containing ice cream and a preparation method thereof. The ice cream is prepared from shrimp protein as a raw material. Due to the addition of the shrimp protein, the ice cream is relatively fine and smooth in mouth feel; due to the addition of lemon juice, the smell of aquatic products can be effectively reduced and the ice cream is fresh in flavor; due to the selective addition of mustard and soy, the specific clean aromas of the aquatic products are brought into full play in the ice cream. The prepared ice cream has the original features of ice cream, is tasty and refreshing as well as sweet and mellow, is relatively soft and natural in color, fine and smooth in mouth feel tasty and variable in flavor, and is relatively good in preservation performance, relatively high in nutritional value and relatively low in unhealthy ingredient content.

(13) Methodology (Details such as; a) variables to be measured and evaluated; b) treatments to be used and their layout c) experimental procedures and design; d) statistical analysis; e) evaluation method and observations to be made, strategies for implementation.)

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Treatments:

Ingredients	T1	T2	T3	T4
Coconut Milk	90 %	80%	70 %	60%
Shrimp broth	10 %	20 %	30 %	40 %

**Other ingredients are consistent.*

Ingredients:

- Shrimp
- Coconut Milk
- Sugar
- Arrowroot
- Fresh milk
- All-purpose Cream
- Salt

Materials:

- Pot
- Measuring cup and spoon
- Weighing Scale
- Mixing bowl
- Blender
- Ladle
- Freezer
- Bowl

Procedures:

Process for minced Shrimp:

1. Clean the shrimp and remove the shell.
2. Set aside the shrimp meat.
3. Add the shrimp shells and 100 ml water into a blender and blend.
4. In a pan, add the shrimp shells and add 200 ml of water. Boil for 15 minutes on low heat and strain then set aside the shrimp broth.
5. Mince the shrimp meat.
6. Cook the mince shrimp in low heat until it becomes flakey

Ice Cream Production;

1. Boil the shrimp broth and add the arrowroot and consistently stirring until thick.
2. In a bowl, combine fresh milk, coconut milk, sugar, and shrimp-arrowroot mixture and mix until all the ingredients mix well.
3. In another bowl, whip all-purpose cream until soft and fluffy. Gradually add the fresh milk and coconut milk mixture and whip until it becomes smooth and creamy.
3. Freeze the mixture into 2 hours and beat slightly again and add mince shrimp.
4. Freeze overnight.

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Experimental Design

This study will make use the experimental method or procedure involving the control or manipulation of conditions for the purpose of study the relative effects of various treatments applied to a number of samples, or of the same treatment applied to a number of different samples. A completely randomized block design (CRD) will be use during the conduct the sensory evaluation.

The Shrimp Ice Cream will carry out four (4) treatments. After processing, the samples will be subjected to sensory, microbial and proximate analysis.

Sensory evaluation

The Shrimp Ice Cream will be prepared in accordance with the protocol for sensory evaluation.

Scorecards that contain the 9-point Hedonic Scale for acceptability will be used to evaluate the development of the product, 30 evaluators will evaluate the product. Only sensory characteristics such as color, taste, flavor, texture, and general acceptability will be evaluated.

For the consumer's preference, 100 respondents of varying ages and sex will evaluate the product.

Proximate and Microbial Analysis

The Shrimp Ice Cream will be brought to the DOST laboratory for proximate composition and microbial analysis.

Data Collection

The data was gathered using a score containing the 9-point hedonic scale. It contains the rating of the panelist on color, taste, flavor, texture, and general acceptability. The data was gathered after every replication and was collected, tallied, tabulated, and computed using mean for acceptability and Analysis of Variance (ANOVA) for the significant difference among treatments.

Scoring Variables

The Criteria for evaluation as to sensory characteristics of the product used 5-point hedonic scale in terms of color, taste, flavor, texture, and general acceptability. The rating for general acceptability in 9- point hedonic scale is 9 for like extremely, 8 for like very much, 7 for like moderately, 6 for like slightly, 5 for neither like nor dislike, 4 for dislike moderately, 3 for dislike slightly, 2 for dislike very much and 1 for dislike extremely (Calmorin 2006).

The score on each item was interpreted based on the result of the computation as to color, taste, flavor, texture, and general acceptability using the following scoring interval and verbal interpretation.

Statistical analysis

The data collected on sensory evaluation was statically analyzed by using SPSS software. The analysis of variance (ANOVA) was used to analyze sensory evaluation results with the significant difference among the sample with different treatments.

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(15) Expected Output *(Indicate the specific products, processes or services which the project aims expected to produce; quantify when possible)*

1. **1 Optimized Shrimp Ice Cream**
2. **1 patented product**
3. **1 IEC material**

(15) Technology Transfer Activity Design *(Indicate the specific products, activity design, processes or services to be transferred as extension activities)*

Research Objectives	Duration (months)				
	1	2	3	4	5
1. Gathering and reading materials connected to the study					
2. Preparation of Proposal					
3. Conduct of Research Study					
4. Analysis of Data gathered from Sensory Observation					
5. Proximate Analysis					
6. Writing the analysis of text and data result					

(16) Targets Beneficiaries *(Who the clienteles are and what are the expected outcome/effects of the use of the projects output)*

1. **Shrimp Farmer**
2. **Coconut Farmer**
3. **Entrepreneurs**
4. **Researchers**
5. **Consumers**

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(17) Personnel Requirement

Name	Role (Program, Project, Study Leader)	Field of Specialization
Stephanie S. Pimentel	Study Leader	Biology
Vialyn B. Ragalado	Research Assistant	Food Science

Percent time devoted to the Program/ Project/ Study

Program/Project Leader (in hours):

Program/Project Members (in hours):

(18) Budgetary Requirement

Ingredients	Quantity	Amount
Shrimp	20kls	3,000.00
Other Ingredients (Fresh Milk, Coconut Milk, Sugar, All Purpose Cream, arrowroot powder, Salt)		5,000.00
Gas during product Development.		1,300.00
Plastic cups for Sensory	5 packs	700.00
Proximate Analysis (moisture, ash, Protein, and Carbohydrates)		10,000.00
Transportation Allowance		1,200.00
Patent		5,000.00
TOTAL		26, 200.00

(19) Literature Cited (include only those references published in reputable journals, and at least that from CAPSU Research Journal,

Balthazar, C. F., Silva, H. . A., Vieira, A. H., Neto, R. P. C., Cappato, L. P., Coimbra, P. T., ... Cruz, A. G. (2017). Assessing the effects of different prebiotic dietary oligosaccharides in sheep milk ice cream. *Food Research International*, 91, 38–46. doi:10.1016/j.foodres.2016.11.008

M. Vedashree • M. R. Asha • C. Roopavati • M. Madhava Naidu Revised: 26 January 2020 / Accepted: 20 March 2020 Association of Food Scientists & Technologists (India) 2020 Characterization of volatile components from ginger plant* at maturity and its value addition to ice cream *J Food Sci Technol* <https://doi.org/10.1007/s13197-020-04370-0>

Research Article: Development of Coconut Milk-Based Spicy Ice Cream as a Nondairy Alternative with Desired Physicochemical and Sensory Attributes K. D. S. S. Perera and O. D. A. N. Perera Department of Food Science and Technology, Faculty of Livestock, Fisheries and Nutrition, Wayamba University of Sri Lanka, Makandura, Gonawila, Sri Lanka. Accepted 23 June 2021; Published 8 July 2021 <https://doi.org/10.1155/2021/6661193>

(20) Capsule Curriculum Vitae (One-page brief C.V. only)